

Adventures in Cognitive Biases



Chapter 1 Beliefs as distributions

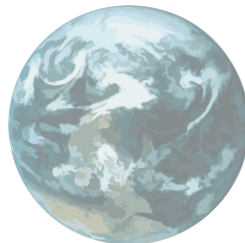
A monk greets you.



Next

MONK: Hello, I have been in search for my successor for a long time. The prophecy says that today is the last possible day for us to meet.

Next



MONK: The world is in danger.

Through the Bystander Effect we have been sitting by the side and watching as a few entities heat up the world and knock systems out of a precarious balance.

Next

The prophecy says that my successor is a tortured soul who will face a perilous journey to rid themselves of 13 biases or so and change the way how others see the world.

13?!?!?

MONK: The prophecy actually thought 88 was more likely, but I thought it would be better to reduce it to 13 or else I'd never find anyone.

But before we worry about the biases of other people, let's work on the ones within ourselves.

Oh dear

Magic Answer Fallacy

The first bias is so pervasive, that it's not formally recognized as a cognitive bias. I call it the **Magic Answer Fallacy**.

It's the tendency to respond to questions with a simple "magic answer".

Next

For instance, how many days are in a year?

Most people would just say 365. But the true length of a year on Earth is ~365.2422 days. That is why we have leap years, but even leap years don't fully account for it.

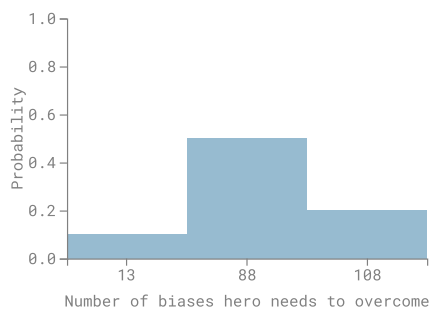
Next

MONK: The first step is to accept that most questions don't have magic answers. Most answers are uncertain, and to capture the uncertainty we draw it as a distribution.

A **distribution** has the possibilities on the x-axis and the probabilities on the y-axis. Consider these distributions from the prophecy.

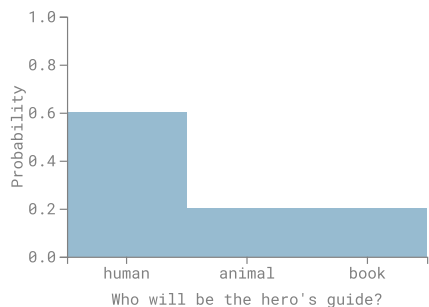
Next

Our hero will likely face a very large number of cognitive biases in their quest... but possibly only 13!



Gee

Our hero is in search of a guide... who will likely be human, but may very well be animal or inanimate.



Okay...

MONK: This point is, distributions allow us to shift from using vague language like "pretty sure", to having a picture of exactly what we think.

Now, begin thinking in distributions instead of magic answers.

Begin

Answer this question to the best of your ability.

Question 1 out of 8. How many hours does an average cat sleep in a day?

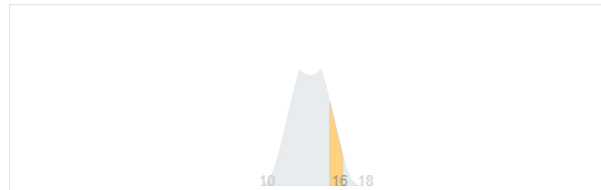
Min guess

Max guess

Next

Drag down the circle so that the distribution covers all answers that you think are reasonable. You can also drag the circle left and right to fine-tune the shape. This diagram is your **belief graph**.

If you assign 0 probability to the magic answer, you will have to do the dishes 🧺.

Submit belief

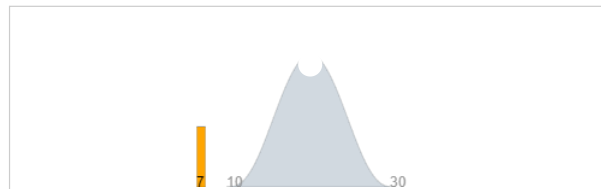
The magic answer was 16. The height of the magic answer on your normalized belief graph was 0.11536.

Onwards

Question 2 out of 8. What percent of total electricity consumption goes to refrigerators?

Min guess

Max guess

NextSubmit belief

The magic answer was 7. The height of the magic answer on your normalized belief graph was 0.

Thereby you will be washing dishes for 1 night 🧺.

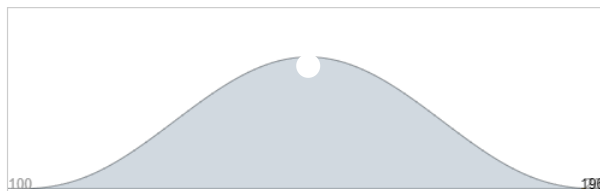
Onwards

Question 3 out of 8. How many countries are there in the world?

Min guess

Max guess

Next



Submit belief

The magic answer was 196. The height of the magic answer on your normalized belief graph was 0.00008.

Onwards

Question 4 out of 8. What percent of the population is left handed?

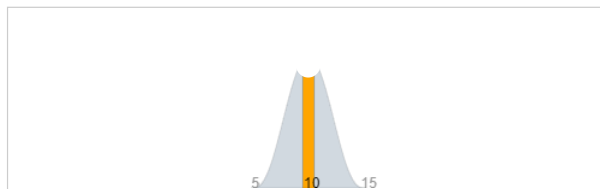
Min guess

Max guess

5

15

Next



Submit belief

The magic answer was 10. The height of the magic answer on your normalized belief graph was 0.21875.

Onwards

Question 5 out of 8. What is the estimated age of the universe in billions of years?

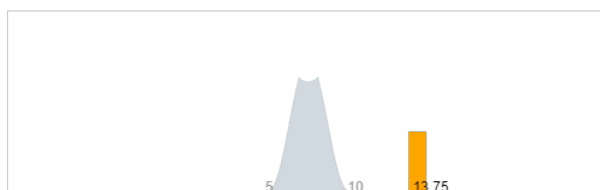
Min guess

Max guess

5

10

Next



Submit belief

The magic answer was 13.75. The height of the magic answer on your normalized belief graph was 0.

Thereby you will be washing dishes for 1 night 🍽️.

Onwards

Question 6 out of 8. What percent of the body's nutrients is consumed by the brain?

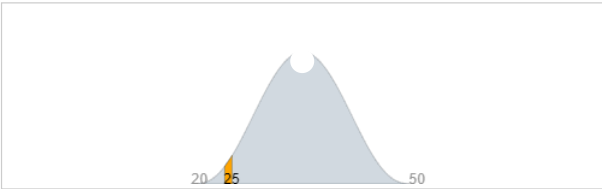
Min guess

Max guess

Next

20

50



Submit belief

The magic answer was 25. The height of the magic answer on your normalized belief graph was 0.0125.

Onwards

Question 7 out of 8. How many bones are there in the human body?

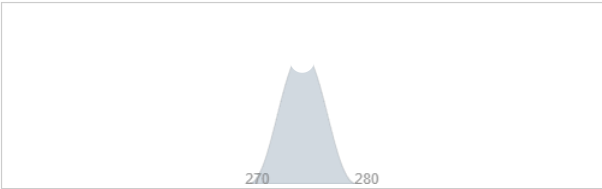
Min guess

Max guess

270

280

Next



Submit belief

The magic answer was 206. The height of the magic answer on your normalized belief graph was 0.

Thereby you will be washing dishes for 1 night 🎁.

Onwards

Question 8 out of 8. How many kilometers tall is Mt. Everest?

Min guess

Max guess

8.5

9

Next



Submit belief